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Logical Diagram
& relational database
schema



ER DIAGRAM
& DATA DICTIONARY



IMPLEMENTATION

Case Study #4:

Research Project Management Office (RMPO)

11 September 2025

Jirapat Datephanyawat (st125843)

Paradorn Khanongsuwan (st125894)

Thet Htun Swe (st126391)

Prepared by group Data Group



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Agenda

Introduction

Project Description

Conceptual Diagram

Relational Database Schema

ER Diagram and Data dictionary

Implementation



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Kookle University Project Initiation Form

Date: 1 Jan 2019
Project Title: Ontology-based Plant Disease Detection System
Project Description: The project aims to ...

Keywords: Ontology, Plant Disease, Intelligent System

Project Category: ☐ Research and Development
☐ Training/Capacity Building
☐ Conference/Workshop

Research Areas of the Project: Semantic Web, Ontology, Knowledge Representation

Project Duration: 12 months

Contract Period: 1 Jan 2019 – 31 Dec 2019

Budget: 1,000,000 THB

Installment:

No.	Date	%Budget	Amount
1	30 April 2019	30	300,000
2	31 August 2019	30	300,000
3	31 December 2019	40	400,000

Principle Investigator:

1. Name: Mark Zuckerberg Email: mark@kmail.com
Department ICT Department Area of Expertise: Social Analytics, AI
2. Name: Elon Musk Email: musk@kmail.com
Department ICT Department Area of Expertise: Knowledge Representation

Project Member:

1. Name: Tim Berners Lee Email: tim@kmail.com
Department ICT Department Area of Expertise: Semantic Web Ontology
2. Name: _____ Email: _____
Department _____ Area of Expertise: _____

Donor/Sponsor Name: Thailand Research Fund

Donor Type: ☐ Research Organization
☐ International Organization
☐ Private Company

Donor Contact Info: Mr. Somchai N. CMMU Building Phayathai Bangkok 10400 Thailand
Tel: (+66) 2 343 1500

Student Assistant:

1. Name: Nina Billa Email: nina@kmail.com
Department: ICT
Responsibility: Programming
Maximum working hours per month: 40 Rate per hour: 300
Start Date: 1 Feb 2019 End date: 30 June 2019
2. Name: Lucy Dell Email: lucy@kmail.com
Department: Business Management
Responsibility: Project Coordinator
Maximum working hours per month: 40 Rate per hour: 300
Start Date: 1 Jan 2019 End date: 31 Dec 2019

Main Focus of the Case Study

This case study focuses on the management of research project data at Kookle University. The RMPO (Research Project Management Office) is responsible for maintaining and tracking information about **research projects, grants, donors, and the faculty, staff, and students** involved.

OBJECTIVE

1. Maintain Data with CRUD Method
2. Analyze and report statistics for management
3. Tracking, planning and forecasting
4. Research performance of each faculty

◀ Spread sheet

Business Rules, Data Maintained and Constraints



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Data Maintained

- **Name:** Project Titles, Donor Names, Staff Names, Student Assistant Names, Department Names, Role Names
- **Identification:** Project ID, Donor ID, Staff ID, Student Assistant ID, Department ID, Installment ID, Keyword ID and Research area ID
- **Financial Data:** Budget Amount, Installment Amount, Student Assistant Rates
- **Classification:** Project Category, Donor Type, Staff Type
- **Date Time:** Contract Start Date, Contract End Date, Installment Dates

Business Rules

- **Budget Management:** Total installments for a project must not exceed the project's total budget amount
- **Resource Allocation:** Staff and student assistants must be properly assigned to departments and projects with consistent tracking of hours and responsibilities
- **Project Classification:** Projects must be categorized as "Research and Development", "Training/Capacity Building", or "Conference/Workshop"
- **Timeline Management:** Contract end dates must be after start dates; project duration is automatically calculated
- **Role Definition:** Staff must be classified as either "Project Member" or "Project Investigator"

Data Constraints

- **Primary Keys:** p_id, d_id, s_id, sa_id, dp_id, i_id, r_id, k_id (**UNIQUE VALUE**) (**AUTO_INCREMENT**)
- **Temporal Constraints:** p_contract_start_date, p_contract_end_date (**Valid datetime format**)
- **Financial Values:** budget_amount, i_amount, sa_rate (**Positive decimal values**)
- **Enumerated Types:**
 - **p_category:** 'Research and Development', 'Training/Capacity Building', 'Conference/Workshop'
 - **d_type:** 'Research Organization', 'International Organization', 'Private Company'
 - **s_role:** 'Project Member', 'Project Investigator'
 - **p_status:** 'Incomplete', 'Complete', 'Ongoing'
- **Contact Information:** d_contact, dp_email, dp_telephone, sa_email, s_email (**Properly formatted**)
- **Calculated Fields:** p_duration (**Generated from contract start/end dates**)



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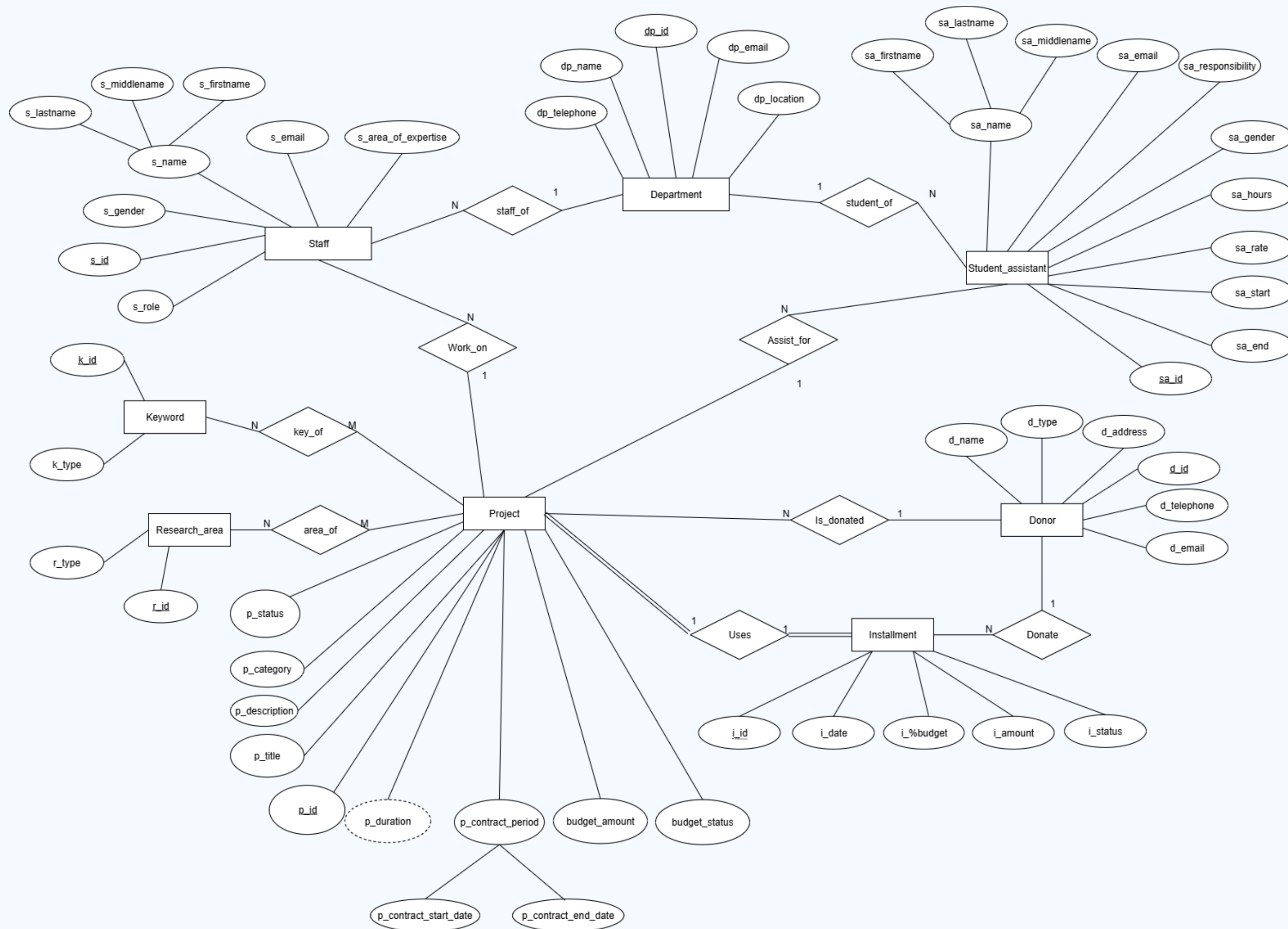
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the conceptual design of our database, which is built to support the RMPO's responsibilities. We've identified eight core tables that are all centered around the Project table.



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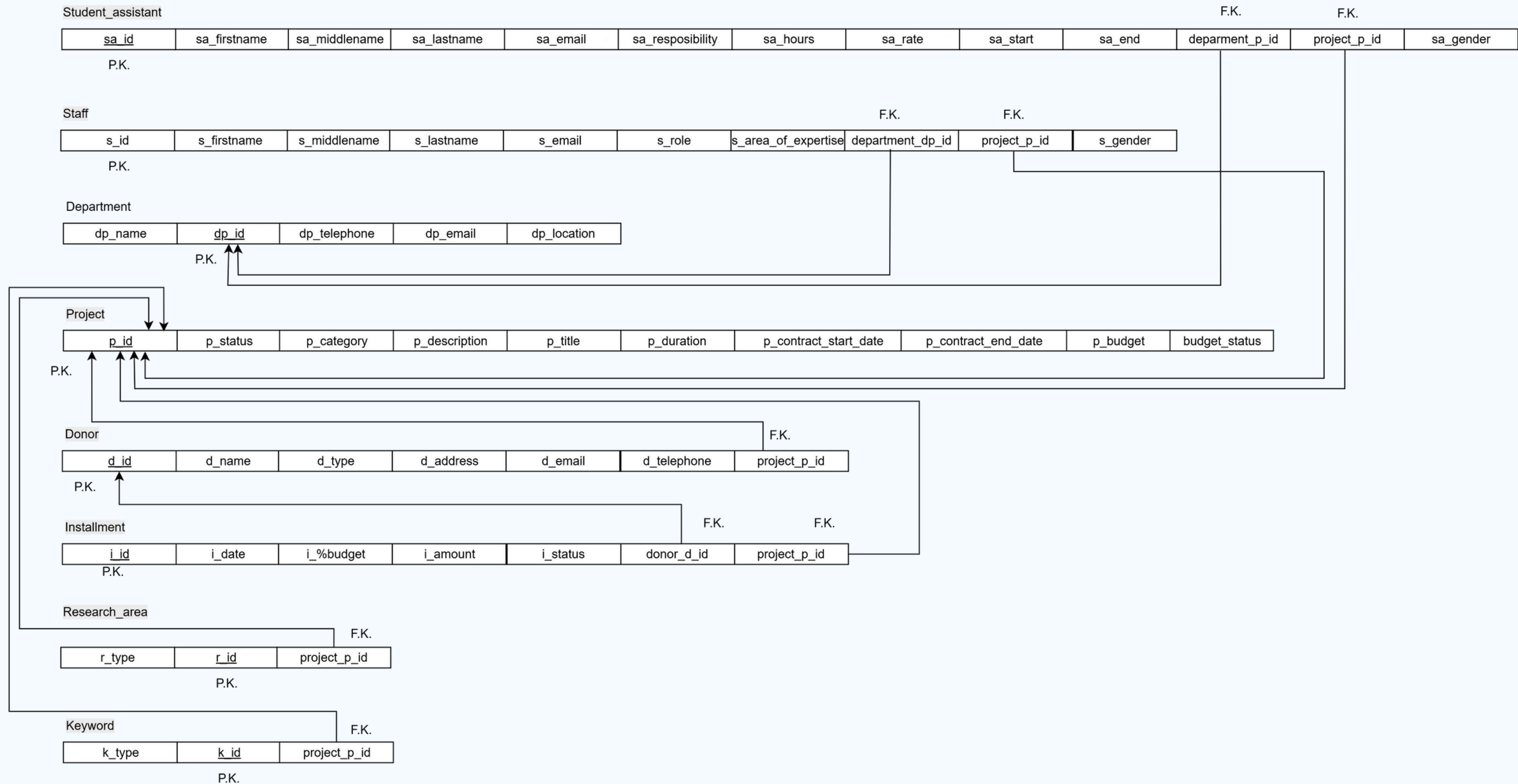
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our Project table is the central hub of our database, with all other tables linking to it through primary and foreign keys to express their importance and relationships.



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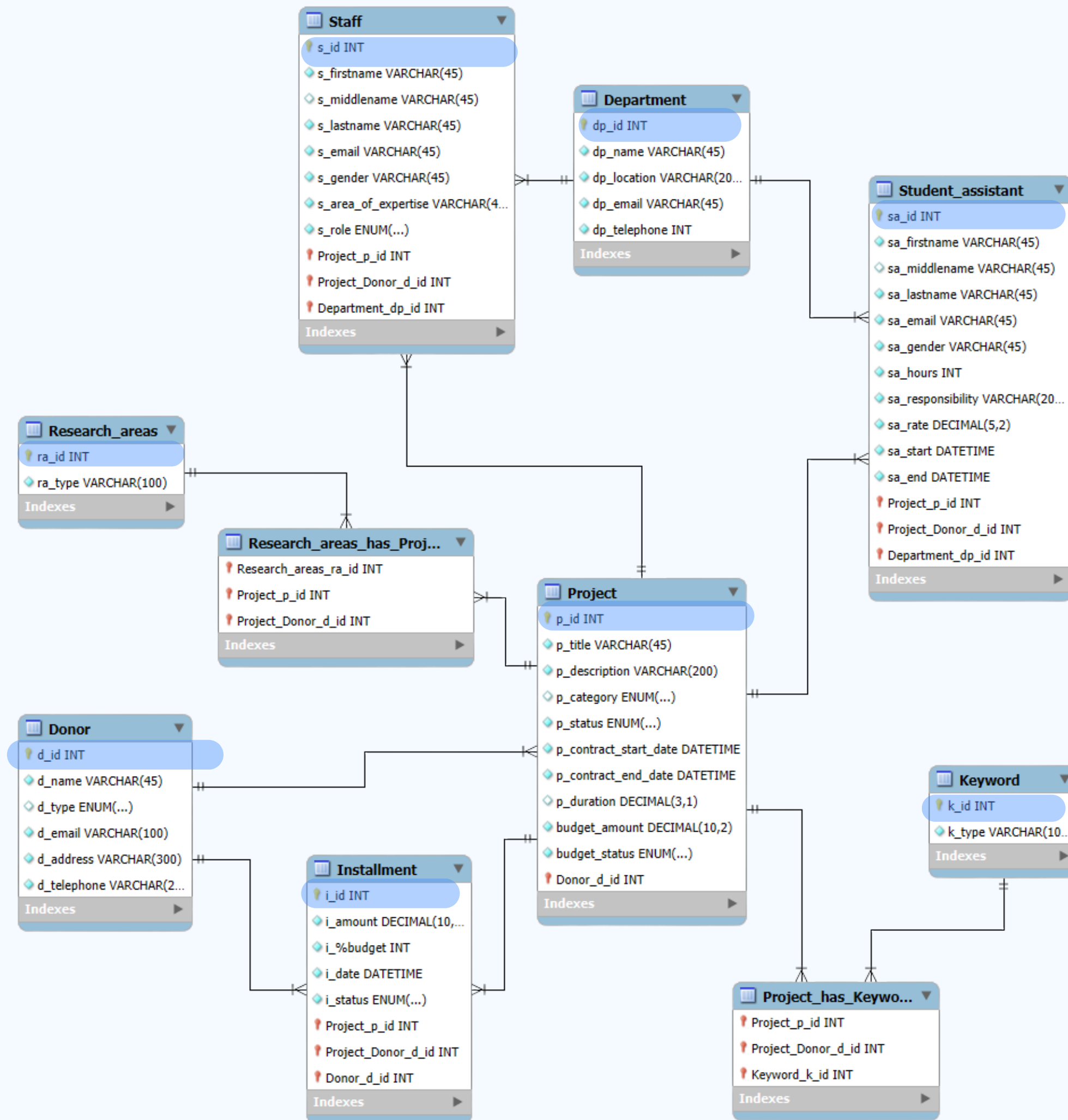
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For all our primary keys, like p_id and s_id, we set them to auto-increment. This allows the database to automatically generate a unique key for each new record. It saves time on data entry and prevents duplicate keys, which keeps our data consistent.



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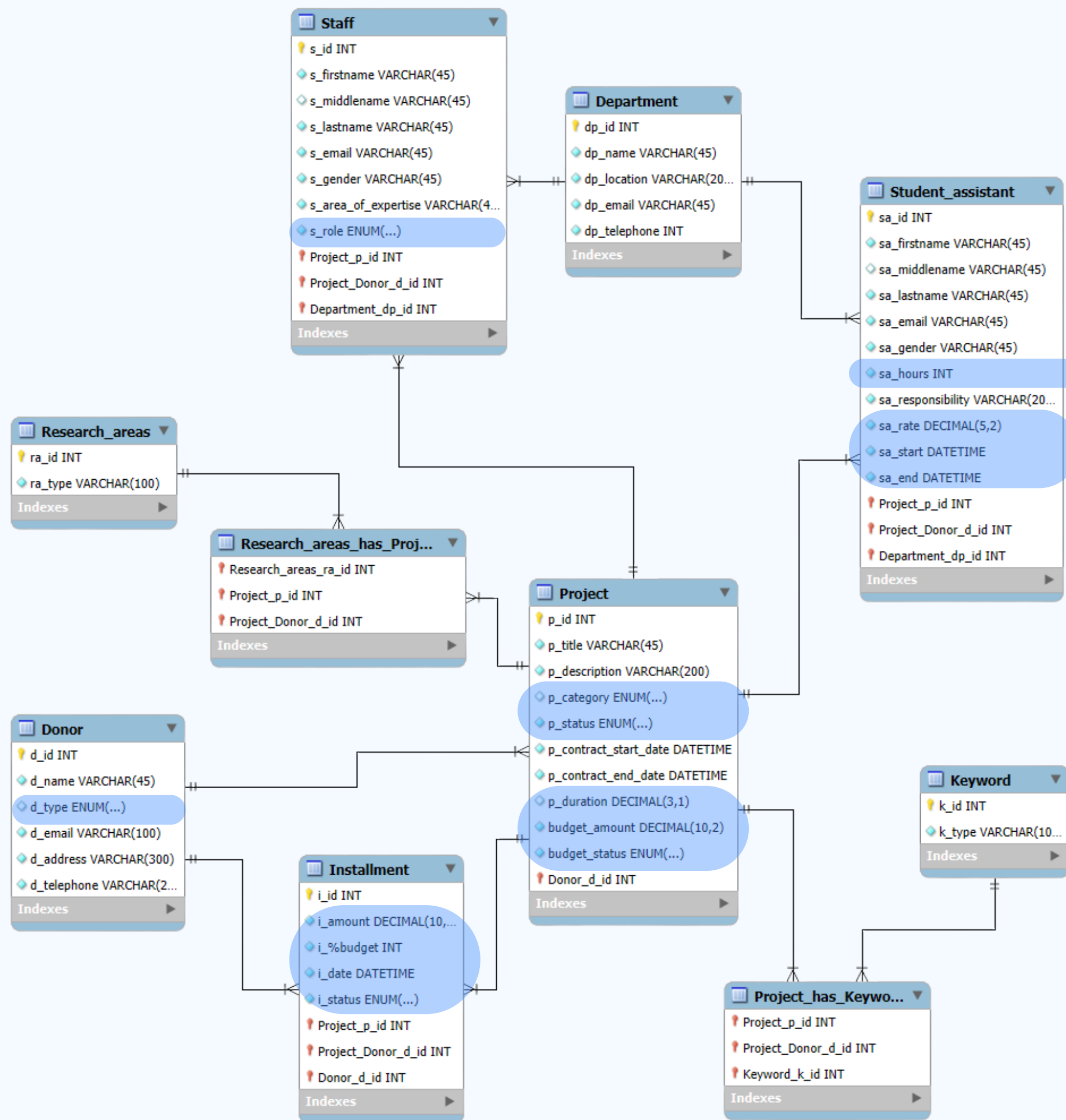
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we defined certain attributes, such as s_role and p_category, as an ENUM data type. This is crucial because it restricts user input to a predefined list of options. For example, p_category can only be 'Research and Development,' 'Training,' or 'Conference.' This ensures data integrity and consistency throughout our database.



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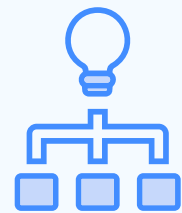
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Scenario theme : Tracking, Planning and Forecasting



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Tracking

1. For all ongoing projects, show the total number of installments, the number of incomplete installments, and the number of complete installments.

```
10  -- T1. For all ongoing projects, show installment counts
11  •  SELECT p.p_id,
12         p.p_title,
13         COUNT(i.i_id) AS total_installments,
14         SUM(CASE WHEN i.i_status = 'Complete' THEN 1 ELSE 0 END) AS complete_installments,
15         SUM(CASE WHEN i.i_status = 'Incomplete' THEN 1 ELSE 0 END) AS incomplete_installments
16  FROM Project p
17  LEFT JOIN Installment i ON p.p_id = i.Project_p_id AND p.Donor_d_id = i.Project_Donor_d_id
18  WHERE p.p_status = 'On going'
19  GROUP BY p.p_id, p.p_title
20  ORDER BY p.p_title;
```

Result Grid

  Filter Rows:

Export: 

Wrap Cell Content: 

	p_id	p_title	total_installments	complete_installments	incomplete_installments
▶	1006	Advanced Robotics for Manufacturing	3	3	0
	1008	Cybersecurity Protocol Development	2	2	0
	1007	Early Childhood Education Models	2	2	0
	1010	Genomic Sequencing of Rare Diseases	2	2	0
	1009	Historical Preservation Techniques	2	2	0
	1011	Microbial Fuel Cells	3	3	0
	1004	Next-Gen Quantum Computing	2	2	0
	1002	Renewable Energy Grid Optimization	2	2	0
	1012	Smart City Traffic Management	2	2	0
	1003	Sustainable Urban Farming	3	3	0

We use a three-step process

- 1) Get the data: The query starts by selecting all projects from the Project table. It uses a LEFT JOIN to connect projects with their installment data from the Installment table. Purpose: This ensures every single project is included, even if it has no installment records yet.
- 2) Filter and Group: It uses a WHERE clause to filter only projects with an 'On going' status. It uses GROUP BY to combine all rows for each project into a single row, which is necessary for the next step.
- 3) Calculate and Display: COUNT(i.i_id): Counts the total number of installments for each project. SUM(CASE WHEN ...): Conditionally counts the completed and incomplete installments. ORDER BY: Sorts the final results by project title for readability.



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IMPLEMENTATION

Tracking

2. Identify projects that have passed their contract end date but are still 'On going'.

```
22  -- T2. Find projects past their end date but still ongoing
23  ●  SELECT p.p_id,
24         p.p_title,
25         p.p_contract_end_date,
26         DATEDIFF(CURDATE(), p.p_contract_end_date) AS days_overdue,
27         d.d_name AS donor_name
28  FROM Project p
29  JOIN Donor d ON p.Donor_d_id = d.d_id
30  WHERE p.p_status = 'On going'
31         AND p.p_contract_end_date < CURDATE()
32  ORDER BY days_overdue DESC;
--
```

	p_id	p_title	p_contract_end_date	days_overdue	donor_name
▶	1006	Advanced Robotics for Manufacturing	2025-08-15 17:00:00	21	National Science Fund
	1002	Renewable Energy Grid Optimization	2025-09-01 17:00:00	4	Government Innovation Agency

We use a three-step process

- 1) The query starts with the Project table (p) and JOIN it with the Donor table (d).
- 2) Filter and Calculate: It uses a WHERE clause to filter the data to include only projects that meet two specific criteria: they must be 'On going' AND their p_contract_end_date must be in the past (< CURDATE()). It then calculates the number of days the project is overdue using DATEDIFF().
- 3) Display the Report: The query selects the project title, contract end date, donor's name, and the calculated days_overdue. It uses ORDER BY with DESC to sort the results, putting the most overdue projects at the top of the list.



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Forecasting

1. For all ongoing projects, calculate the total estimated budget to be spent in each year (based on the contract end date).

```
38  -- F1. Budget spending by year for ongoing projects
39  ●  SELECT YEAR(p.p_contract_end_date) AS forecast_year,
40         COUNT(p.p_id) AS projects_ending,
41         SUM(p.budget_amount) AS total_budget
42  FROM Project p
43  WHERE p.p_status = 'On going'
44  GROUP BY YEAR(p.p_contract_end_date)
45  ORDER BY forecast_year;
--
```

Result Grid				Filter Rows:		Export:	Wrap Cell Content:
	forecast_year	projects_ending	total_budget				
▶	2025	4	11500000.00				
	2026	5	13200000.00				
	2027	1	3000000.00				

We use a two-step process

- 1) Filter Projects: We use the WHERE clause to filter the Project table and select only those projects that are currently 'On going'.
- 2) Calculate and Group: The query then uses aggregate functions to calculate the forecast_year from the p_contract_end_date, count the number of projects ending in each year (projects_ending), and sum their budgets (total_budget). The results are grouped by the forecast_year.



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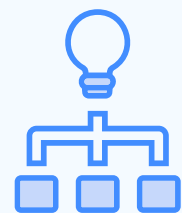
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Forecasting

2. For each ongoing project, show the project lead (Principal Investigator), the total number of staff, and the project budget, to help in planning staff for new projects with similar budgets.

```
47 -- F2. Project leads and team sizes for ongoing projects
48 • SELECT p.p_id,
49         p.p_title,
50         p.budget_amount,
51         CONCAT(s.s_firstname, ' ', s.s_lastname) AS principal_investigator,
52         staff_count.total_staff
53 FROM Project p
54 LEFT JOIN Staff s ON p.p_id = s.Project_p_id
55                   AND p.Donor_d_id = s.Project_Donor_d_id
56                   AND s.s_role = 'Principle Investigator'
57 LEFT JOIN (
58     SELECT Project_p_id, Project_Donor_d_id, COUNT(*) AS total_staff
59     FROM Staff
60     GROUP BY Project_p_id, Project_Donor_d_id
61 ) staff_count ON p.p_id = staff_count.Project_p_id AND p.Donor_d_id = staff_count.Project_Donor_d_id
62 WHERE p.p_status = 'On going'
63 ORDER BY p.budget_amount DESC;
```

	p_id	p_title	budget_amount	principal_investigator	total_staff
▶	1008	Cybersecurity Protocol Development	4800000.00	Daniel Miller	4
	1004	Next-Gen Quantum Computing	4200000.00	John Smith	4
	1002	Renewable Energy Grid Optimization	3500000.00	David Lee	4
	1010	Genomic Sequencing of Rare Diseases	3200000.00	Laura Evans	4
	1012	Smart City Traffic Management	3000000.00	Megan Hall	4
	1011	Microbial Fuel Cells	2800000.00	Robert Clark	4
	1006	Advanced Robotics for Manufacturing	2100000.00	Michael Brown	4
	1003	Sustainable Urban Farming	1800000.00	Maria Garcia	4
	1007	Early Childhood Education Models	1200000.00	Jessica Wilson	4
	1009	Historical Preservation Techniques	1100000.00	Olivia Moore	4

We use a two-step process

- 1) Count Staff: The query starts with a subquery that counts the total number of staff members for each project.
- 2) Combine and Select Data: The main query then joins the Project and Staff tables with the results from the staff count subquery. This allows it to show the project title, budget, Principal Investigator's name, and the total number of staff for each ongoing project. The WHERE clause filters for projects that are On going, and the results are ordered by budget amount.



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Planning

1. Identify the areas of expertise with the fewest staff members in ongoing projects, for future recruitment planning.

```
69  -- P1. Areas of expertise with few staff in ongoing projects
70  •  SELECT s.s_area_of_expertise,
71         COUNT(s.s_id) AS staff_count
72  FROM Staff s
73  JOIN Project p ON s.Project_p_id = p.p_id AND s.Project_Donor_d_id = p.Donor_d_id
74  WHERE p.p_status = 'On going'
75  GROUP BY s.s_area_of_expertise
76  ORDER BY staff_count ASC;
--
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

	s_area_of_expertise	staff_count
▶	Genomic Sequencing	1
	Cybersecurity	1
	Network Security	1

We use a three-step process

- 1) Select and Combine Data: The query uses a SELECT statement to get s_area_of_expertise and the COUNT of s_id as staff_count from the Staff and Project tables. The JOIN command links the tables together based on Project_p_id.
- 2) Filter and Group: The WHERE clause filters the results to only include projects that are 'On going'. The GROUP BY clause then aggregates the staff count for each unique s_area_of_expertise.
- 3) Order the Results: Finally, the ORDER BY clause sorts the areas of expertise by their staff count in ascending order (ASC), showing the areas with the fewest staff first.



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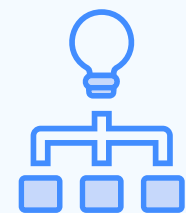
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Planning

2. Identify available staff members (not currently part of an ongoing project) who can be assigned to new projects.

```
79 • SELECT DISTINCT s.s_id,
80         CONCAT(s.s_firstname, ' ', s.s_lastname) AS staff_name,
81         s.s_area_of_expertise,
82         d.dp_name AS department
83 FROM Staff s
84 JOIN Department d ON s.Department_dp_id = d.dp_id
85 JOIN Project p ON s.Project_p_id = p.p_id AND s.Project_Donor_d_id = p.Donor_d_id
86 WHERE p.p_status = 'Complete'
87 AND s.s_id NOT IN (
88     SELECT s2.s_id
89     FROM Staff s2
90     JOIN Project p2 ON s2.Project_p_id = p2.p_id AND s2.Project_Donor_d_id = p2.Donor_d_id
91     WHERE p2.p_status = 'On going'
92 )
93 ORDER BY s.s_area_of_expertise, staff_name;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

	s_id	staff_name	s_area_of_expertise	department
▶	3	David Lee	Algorithm Development	Engineering
	2	Laura Evans	Data Science	Research
	19	Olivia Moore	Ecosystem Modeling	Administration
	1	Alex Chen	Machine Learning	Research
	17	Emily Davis	Marine Biology	Administration
	20	Rebecca Thomas	Marine Conservation	Sciences
	18	Nancy Phillips	Oceanography	Sciences
	4	Michael Brown	Software Engineering	Engineering

We use a two-step process

- 1) Find Staff on Ongoing Projects: A subquery is first executed to find all staff members who are currently assigned to a project with the status 'On going'.
- 2) Identify Available Staff: The main query then filters the entire staff list, excluding anyone found in the first step. This provides a list of staff members who are not currently part of an active project.



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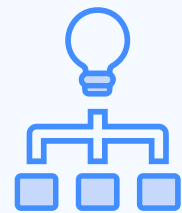
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Thank you